

## Trainings ▾

### Install

#### with Mechanically Actuated Injector

Do **not** lubricate the o-rings until the injector is ready for installation in the cylinder head.

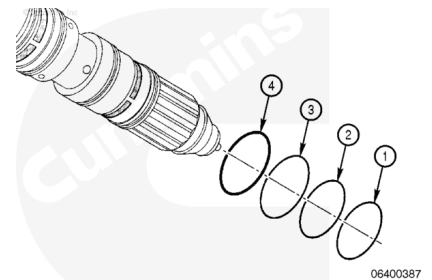
Install four new o-rings on the injector. Do **not** twist the o-rings.

O-ring (1) is brown and is smaller in diameter than the other three.

O-ring (2) is green.

O-ring (3) is blue and is larger in diameter than (2), but has a smaller cross section than o-ring (4).

O-ring (4) is black and is the largest in diameter and cross section.

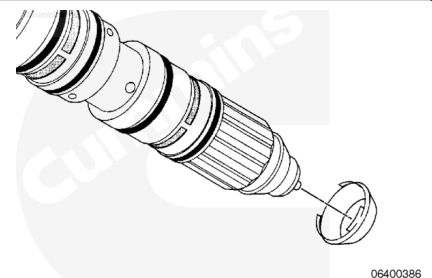


Install the proper size sealing ring.

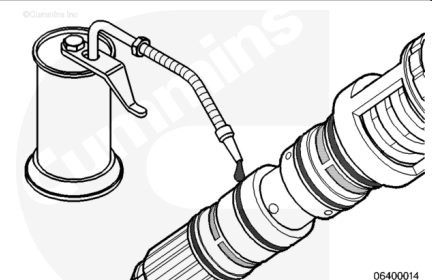
Refer to Procedure 002-004

(</qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html>) in Section 2 for instructions on how to measure injector protrusion and select the correct seal.

**Note : Incorrect injector protrusion can lead to premature injector tip wear.**



Use clean engine lubricating oil (15W-40) to lubricate the o-rings.



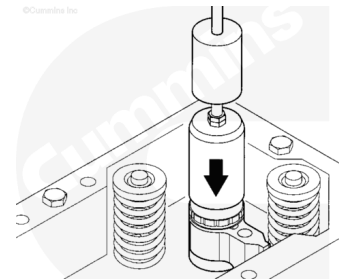
Install the hold-down clamp on the side of the injector.

Install the injector and hold-down clamp in the cylinder head.

**Note :** If the injector installation tool, Part Number 3824830, is not available, use a 40 mm [1 9/16 inch] socket with an extension and a rubber mallet to install the injector.

Use injector installation tool, Part Number 3824830, to set the injector in the bore.

The slide hammer will make a dull sound when the injector is seated properly.



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Use clean engine oil to lubricate the injector hold-down capscrew.

Install the new washer and injector hold-down capscrew.

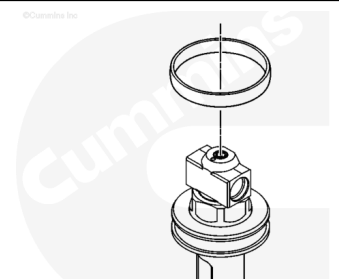
**Torque Value:** 75 n•m [ 55 ft-lb ]



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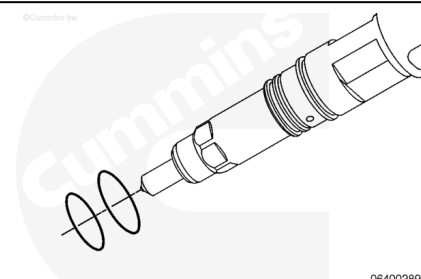
### with Electronically Actuated Injector

Install the new o-ring/damper on the injector.



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Install the new o-rings on the injector body.

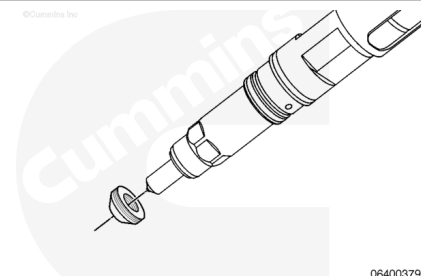


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### **⚠ CAUTION ⚠**

**DO NOT** reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

Install the new seal on the end of the injector.



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**⚠ WARNING ⚠**

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

**⚠ WARNING ⚠**

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

**⚠ WARNING ⚠**

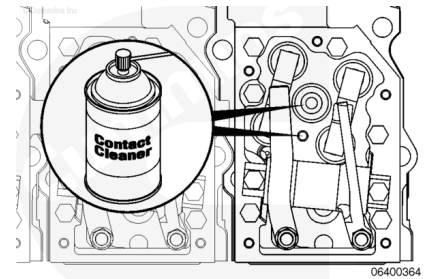
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

**⚠ CAUTION ⚠**

Keep the injector bore and combustion chamber protected when using compressed air to prevent contamination. Failure to do so can cause engine damage.

**⚠ CAUTION ⚠**

**DO NOT** reuse the injector hold down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

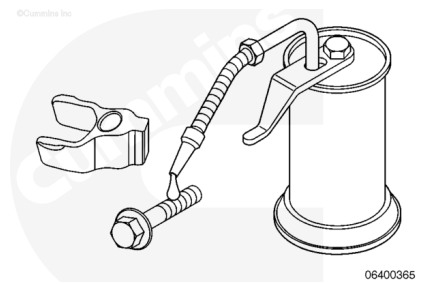


**Note :** On applications with high drain line head pressure, such as a fuel tank mounted higher than the engine, it is possible for fuel to leak from the drain line into the cylinder when the injector is removed. Prior to installing the injector, check for fuel dripping into the cylinder and evacuate the fuel before installing the injector.

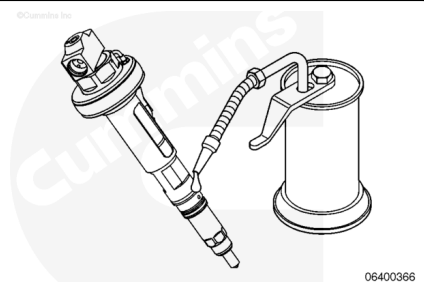
Clean the injector bore in the cylinder head with electrical contact cleaner, Part Number 3824510, or equivalent, to make sure the o-rings will seal properly. Verify the injector bore is free of debris to allow proper sealing of the new o-rings.

Clean the threaded hole in the cylinder head where the injector hold-down capscrew will be installed with electrical contact cleaner, Part Number 3824510, or equivalent, and compressed air.

Lubricate the threads and underside of the new injector hold-down capscrew with clean engine lubricating oil.

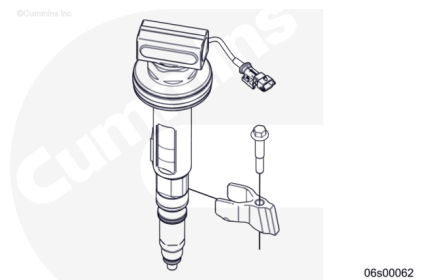


Lubricate all the o-rings with a light coating of clean engine lubricating oil.



**⚠ CAUTION ⚠**

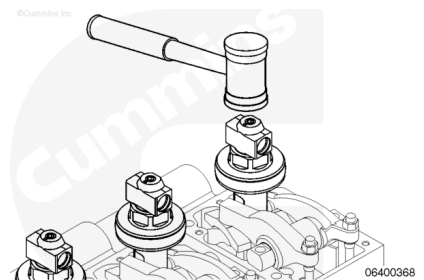
**Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.**



Assemble the injector hold-down clamp and capscrew into the injector.

**⚠ CAUTION ⚠**

**Only use a soft face mallet. Otherwise damage to the injector can occur.**



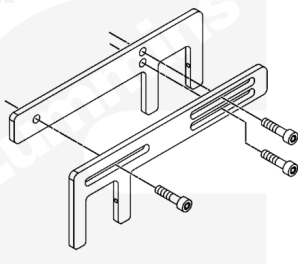
The injector installer can be used on single or multiple injector installations.

Install the assembly into the cylinder head by tapping gently on the injector with a soft face mallet.

Verify the injector is fully seated in the cylinder head before proceeding.

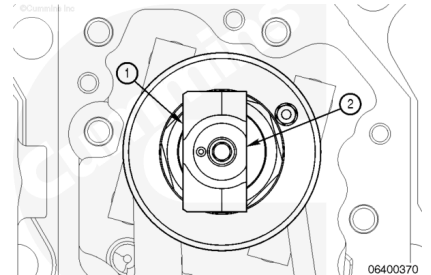
**Note :** If the injector is not fully seated prior to torquing it can lead to engine damage during operation.

Assemble the injector installer kit, Part Number 4918620.  
Finger-tighten the three shoulder capscrews.



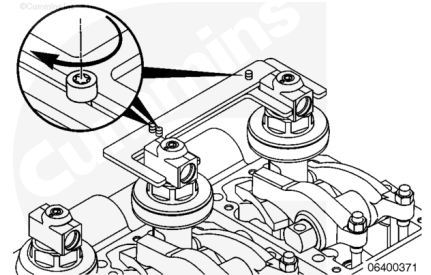
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Place the injector installer on the injector such that the ball spring plunger is on the tapered side of the injector (1). If necessary, adjust the ball spring plunger so that it presses against the injector. Make sure the flat side of the injector (2) sits flush with the injector installer.



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Align the injector installer with two injectors.



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Verify the injector hold-down clamp is properly located.

Hand tighten the injector hold-down clamp capscrew two to three turns to start, verifying the hold-down clamp capscrew engages properly to prevent cross threading.

Torque the injector hold-down capscrews.

The 60 degree torque angle can be achieved at the capscrew by marking the location of one corner of the capscrew to a location on the injector hold-down clamp.

Tighten to:

**Torque Value:** 68 n•m [ 50 ft-lb ]

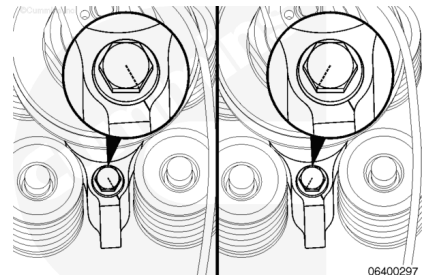
Back off completely.

Tighten to:

**Torque Value:** 30 n•m [ 22 ft-lb ]

Rotate the capscrew 60 degrees.

Remove the injector installer.



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